Voron V2 Bed Level Camera

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VIEW IN BROWSER

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Summary

In case you don't want the bird's eye view

3D Printers > 3D Printers - Upgrades

Tags: camera mod voron webcam

This is a case and extrusion mount for inexpensive 1080p webcams. It's just low enough to clear the V2 gantry. Tested with Xol2 non-UHF hotend without kinematic mounts. If you use this successfully with other hotend combinations, please let me know and I'll update the description. If you use a kinematic mount, let me know what the difference is and I can make an additional extrusion mount for that height.

BOM

- 1x M3 20mm BHCS
- 3x M2 10mm self-tapping screws
- 1x 3x5x4 “Voron standard” heatset
- 1x Extrusion nut (roll-in, T-nut, etc.)
- 1x Small zip tie
- 1x Generic 1080p webcam (eMeet C960 is what I used but I suspect most webcams are similar without the case)

Instructions

1. Print 1 of each part
2. Insert heat set into Extrusion Mount
3. Open the webcam case (see below)
4. Once just the PCB is isolated, inspect the lens. Most of these cameras have a threaded (screw on) lens with a dab of glue. If so, twist the lens to break the glue seal. Rotate the lens so that it's about a quarter turn out(to the left) from it's original position. This step is necessary as most webcams are set with the focus distance too far away.
5. Insert the PCB into the Back Plate.
6. Put the Front Cover over top of the Back Plate, holding both in place with one hand.
7. Thread the M2 screws in from the back.
8. Thread the M3 screw through the bottom of the Back Plate, and fasten to the Extrusion Mount.
9. Install extrusion not into 2020 frame and screw Extrusion Mount in place
10. Adjusting the pivot direction as needed
11. Place an item in the middle of the build plate to check how focus is. If everything is out of focus, twist the lens in (to the right). If the background is in focus but the printed object is not, twist the lens out (to the left).
12. Add a zip tie to secure the USB cable
13. Route the cable into the basement using your preferred method (belt covers with cable routing slots help here)
14. **IMPORANT LAST STEP: Slowly lower the gantry down, then move the Y axis back and forth to make sure everything clears before running a print.** I tested the model with a Xol2 toolhead with a Rapido HF (not the longer UHF), but your toolhead and hotend combination may vary slightly, so be sure to check!

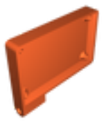
Opening the eMeet C960 Case

1. Where the main case pivots from the mount, there are two rubber dots. Use an awl, ice pick, or thin need nose pliers to pry these out.
2. Remove the pivot screws.
3. To remove the case, there's a small flat dimple that a flathead screwdriver will fit on the bottom. Use the screwdriver to pry it apart. This will break the case irreparably.
4. Once the case is halved, use plyers to pull the case apart to loosen the PCB. The case material is flexible and will bend enough to allow the PCB to easily come out. Be careful not to damage the PCB in the process.

Model files

extrusion-mount-no-kinematics.stl

front-cover.stl



back-plate.stl

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